Lipoprotein Particle Profiles by Nuclear Magnetic Resonance Compared with Standard Lipids and Apolipoproteins in Predicting Incident Cardiovascular Disease in Women

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Background Nuclear magnetic resonance (NMR) spectroscopy measures the number and size of lipoprotein particles, instead of their cholesterol or triglyceride content, but its clinical utility is uncertain. Methods and Results Baseline lipoproteins were measured by NMR in 27,673 initially healthy women followed for incident cardiovascular disease (CVD, N=1,015) over 11 years. Adjusting for non-lipid risk factors, hazard ratios (HRs) and 95% confidence intervals (CIs) for top vs bottom quintile of NMR-measured lipoprotein particle concentration (particles/L) were, for low-density lipoprotein (LDLNMR) 2.51 (1.91–3.30), high-density lipoprotein (HDLNMR) 0.91 (0.75–1.12), very-low-density lipoprotein (VLDLNMR) 1.71 (1.38–2.12), and LDLNMR/HDLNMR ratio 2.25 (1.80–2.81). Similarly-adjusted results for NMR-measured lipoprotein particle size (nanometers) were, for LDLNMR size 0.64 (0.52–0.79), HDLNMR size 0.65 (0.51–0.81), and VLDLNMR size 1.37 (1.10–1.70). Hazard ratios for NMR measures were comparable but not superior to standard lipids: total cholesterol 2.08 (1.63–2.67), LDL cholesterol 1.74 (1.40–2.16), HDL cholesterol 0.52 (0.42–0.64), triglycerides 2.58 (1.95–3.41), non-HDL cholesterol 2.52 (1.95–3.25), total/HDL cholesterol ratio 2.82 (2.23–3.58); and apolipoproteins: B100 2.57 (1.98–3.33), A-1 0.63 (0.52–0.77), B100/A-1 ratio 2.79 (2.21–3.54). There was essentially no reclassification improvement with adding LDLNMR particle concentration or apolipoprotein B100 to a model that already included the total/HDL cholesterol ratio and non-lipid risk factors (net reclassification index [NRI], 0% and 1.9%, respectively), nor did the addition of either variable result in a statistically significant improvement in the c-index. Conclusions In this prospective study of healthy women, CVD risk prediction associated with lipoprotein profiles evaluated by NMR was comparable but not superior to standard lipids or apolipoproteins.